

RedLine Stealer

Comprehensive Malware Analysis

A SOC Analyst & Malware Analyst Perspective
From Threat Intelligence to Layered Defense

PROJECT INTRODUCTION | 2026

Project Overview

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Introduction & Threat Landscape

Understanding RedLine Stealer, its distribution mechanisms, and the evolving information-stealer threat landscape

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A structured ten-task analytical framework covering the complete malware analysis lifecycle

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Threat intelligence profiling, lab environment design, and static malware analysis

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Behavioral analysis, credential theft examination, and IOC extraction

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01

Introduction & Threat Landscape

Understanding RedLine Stealer, its distribution mechanisms, and the evolving information-stealer threat landscape

RedLine Stealer — Threat Profile

RedLine Stealer is one of the most actively distributed **information-stealing malware** families. It operates as a **Malware-as-a-Service (MaaS)**, accessible to threat actors of all skill levels.

Distributed via cracked software, pirated games, and fraudulent installers, it **silently harvests sensitive data** from compromised endpoints.

Stolen data is serialized into **XML** and exfiltrated via **SOAP-based C2 messaging**, blending with legitimate web traffic to evade detection.

Target Data

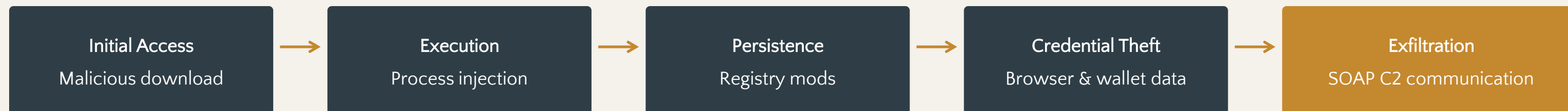
Browser credentials, cookies, cryptocurrency wallets, session tokens from Discord, Telegram, Steam, and VPN clients

Distribution Vectors

Cracked software, pirated games, fraudulent installers, and spam campaigns via email and messaging platforms

KEY CHALLENGE | SOAP-based C2 exfiltration blends with legitimate web traffic, making RedLine particularly difficult to detect with conventional network monitoring tools. This project addresses that gap through comprehensive analysis and engineered detection.

The Infection Chain & Attack Vectors



Infection Chain Analysis

The infection chain progresses from **initial delivery** to **exfiltration** in five stages:

1. Victim executes a malicious payload disguised as legitimate software
2. RedLine injects into processes to establish persistence
3. Malware harvests credentials, cookies, and wallet data

4. Data is serialized into XML for structured transmission

5. Exfiltrated via SOAP messages to C2, masquerading as legitimate traffic

SOAP-based exfiltration is a key differentiator — it bypasses traditional network detection relying on anomalous protocols or ports.

Project Objective: Bridge the detection gap by combining offensive malware understanding with defensive engineering for SOC teams.

02

Project Methodology

A structured ten-task analytical framework covering the complete malware analysis lifecycle — from threat intelligence to prevention strategy

Analysis Framework — Tasks 1 to 3

01 Threat Intelligence

Establishes foundational knowledge covering RedLine's classification, capabilities, and infection chain.

Key outputs:

Threat actor profile, TTP mapping, attack chain documentation

02 Lab Design

Isolated environment for safe execution:
Windows victim with Wazuh SIEM monitoring.

Key outputs:

Isolated network, Wazuh server, victim VM

03 Static Analysis

Pre-execution examination: hash extraction (MD5, SHA1, SHA256), PE header analysis, strings extraction.

Key outputs:

File hashes, PE structure, VirusTotal correlation

PHASE OBJECTIVE | Establish a complete understanding of the threat and a safe, monitored environment for controlled malware execution and analysis.

Deep Analysis — Tasks 4 to 6

04 Behavioral Analysis

Captures runtime behavior: process creation, file system mods, registry changes, network connections.

Key outputs:

Injection patterns, persistence mechanisms, C2 analysis

05 Credential Theft Analysis

Examines targeted browsers, stolen data types, and crypto wallet harvesting methods.

Key outputs:

Targeted app list, data taxonomy, wallet methodology

06 IOC Extraction

Compiles all IOCs into a structured database ready for detection tool deployment.

Key outputs:

Hashes, paths, registry keys, domains, IPs, mutexes

PHASE OBJECTIVE | Understand the malware's runtime behavior, data theft capabilities, and extract actionable indicators for defensive operations.

Defense Operations — Tasks 7 to 9

07 Detection Engineering

Detection rules for browser data access, credential dumping, and PowerShell abuse.

Key outputs:

Sigma rules, behavioral detections, C2 alerts

08 Elastik SIEM Integration

Elastic configuration for RedLine monitoring with custom alert rules.

Key outputs:

Process alerts, registry monitoring, network tracking

09 Threat Hunting

Incident response case study: alert triage, process ID, IOC correlation, containment.

Key outputs:

IR playbook, triage workflow, containment procedures

PHASE OBJECTIVE | Transition from analysis to active defense — building detection capabilities and simulating real-world SOC investigation workflows.

Prevention Strategy – Task 10

Layered Defense Framework

Email Security

Gateways and attachment filtering to block malicious payloads at entry

Browser Hardening

Credential isolation and secure storage to prevent harvesting

MFA Deployment

Multi-factor authentication to protect stolen credentials from reuse

EDR Deployment

Behavioral monitoring and endpoint detection for real-time response

User Awareness Training

Phishing simulation and security education to reduce social engineering success

Patch Management

Software integrity verification and timely patching to close exploitation vectors

Network Monitoring

Outbound connection analysis and DNS monitoring to detect C2 activity

DEFENSE IN DEPTH | Each control is directly mapped to a specific RedLine TTP. No single control is sufficient — the layered approach ensures multiple opportunities to detect and block the threat across the kill chain.

Project Summary & Key Outcomes

This project delivers **10 comprehensive analytical tasks** spanning the complete malware analysis lifecycle — from red-team understanding to blue-team defense — producing actionable intelligence for security operations teams.

Task	Focus Area	Key Deliverable
1	Threat Intelligence	Threat actor profile, TTP mapping, attack chain documentation
2	Lab Design	Isolated analysis environment with Wazuh SIEM monitoring
3	Static Analysis	Hash extraction, PE analysis, strings, VirusTotal correlation
4	Behavioral Analysis	Runtime behavior report: processes, registry, network, files
5	Credential Theft	Targeted application list, stolen data taxonomy, wallet analysis
6	IOC Extraction	Structured IOC database: hashes, paths, keys, domains, IPs
7	Detection Engineering	Sigma rules and behavioral detections for SIEM deployment
8	Wazuh SIEM Integration	Configured Wazuh with custom alert rules for RedLine activity
9	Threat Hunting	Incident response playbook with triage and containment workflow
10	Prevention Strategy	Layered defense framework mapped to RedLine TTPs

Core Value: Bridges offensive malware understanding with defensive application — every deliverable is designed for direct deployment by SOC teams.

Thank You

REDLINE STEALER MALWARE ANALYSIS PROJECT

A comprehensive analysis bridging offensive understanding
with defensive application for SOC operations

QUESTIONS & DISCUSSION